

Experiment No: 01

Experiment Name: Study the Role of Power Electronics for Environment and Sustainability.

Abstract:

Power electronics is the technology that converts, controls, and regulates electrical power using semiconductor switching devices, and it has become one of the most important enablers of an efficient, reliable, and sustainable energy system. This report examines the role of power electronics in environmental sustainability by discussing its fundamental features, key applications, recent technological developments, market trends, future prospects, and its growing significance in Bangladesh. Power-electronic converters and inverters are central to integrating renewable energy sources such as solar and wind into modern power systems, improving the efficiency of industrial motor drives, supporting energy storage, and enabling the adoption of electric vehicles. Recent advances in wide-bandgap semiconductors, particularly silicon carbide (SiC) and gallium nitride (GaN), have pushed converter efficiencies beyond 98 percent [9]–[11]. In Bangladesh, renewable energy currently accounts for only a small share of total power generation; however, national policy targets and real-time grid data analyzed in this report indicate that power-electronic technologies will be essential for expanding solar capacity, stabilizing the grid, and meeting the country's 2030 and 2041 renewable-energy targets.

Introduction:

Climate change and the continuous growth of global electricity demand have made efficient energy generation, transmission, and utilization one of the greatest engineering challenges of this century. Power electronics — the conversion and control of electrical power using devices such as diodes, thyristors, MOSFETs, and IGBTs — has become central to addressing this challenge, since nearly every stage of a modern power system now passes through some form of electronic conversion [2]. Beyond traditional motor drives and uninterruptible power supplies, power electronics is now essential for integrating renewable sources, electric vehicles, HVDC transmission, and smart grids, all of which depend on converters for voltage regulation, frequency control, and grid synchronization [3].

This report presents a study of the fundamental features of power electronics, its contribution to renewable-energy integration, recent developments in wide-bandgap semiconductors, global market trends, and the current situation and future prospects in Bangladesh, supported by up-to-date generation and demand data obtained from the Power Grid Company of Bangladesh (PGCB).

Key Features of Power Electronics for Sustainability:

Power-electronic converters transfer electricity from one stage of the power system to the next while allowing its voltage, frequency, and waveform to be flexibly controlled. This flexibility underlies several features that make power electronics central to environmental and energy sustainability:

- Renewable integration: Inverters convert DC output from solar photovoltaic panels and battery systems, and back-to-back converters regulate variable-frequency wind power, into stable grid-compatible AC [3].
- Wide-bandgap devices: Silicon carbide (SiC) and gallium nitride (GaN) semiconductors switch faster and operate at higher voltage and temperature than conventional silicon, cutting conversion losses and reaching efficiencies close to 99 percent [9]–[11].
- Smart-grid and HVDC integration: Power-electronic interfaces allow bidirectional power flow, voltage regulation, and grid-stability support as more variable renewable capacity is added.
- Electrification of transport: Traction inverters, on-board chargers, and DC–DC converters manage power flow between batteries, motors, and charging systems in electric and hybrid vehicles.
- Green industry and motor drives: Variable Frequency Drives (VFDs) reduce unnecessary energy consumption in industrial motors, pumps, and compressors.
- Energy storage and microgrids: Bidirectional converters regulate battery charging and discharging, supporting grid stability and enabling islanded microgrid operation during disturbances.

Data:

This section presents recent quantitative data on the global power-electronics market, converter efficiency, and the current operating condition of the Bangladesh national grid. Grid data were obtained directly from the Power Grid Company of Bangladesh (PGCB) ERP portal; market and technology figures are drawn from industry reports cited in IEEE format.

The global power-electronics market for renewable-energy applications was valued at approximately USD 16–18 billion in 2025 and is projected to grow at 8–9 percent CAGR, exceeding USD 40 billion by the mid-2030s [13]–[15]. Figure 1 illustrates this projected growth trend.

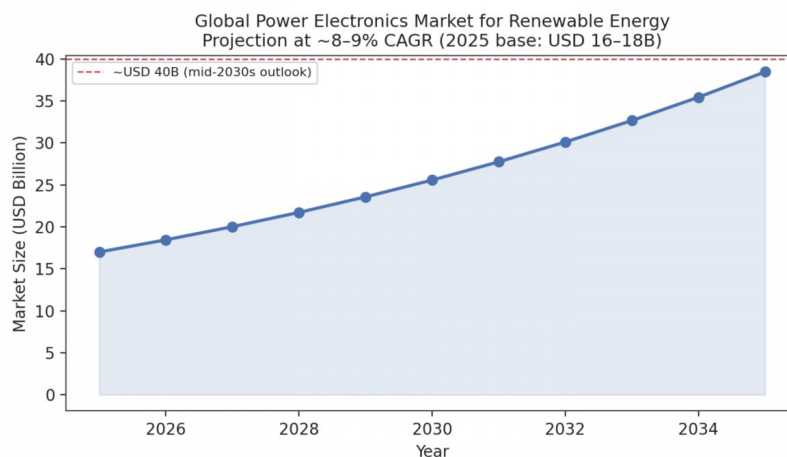


Figure 1. Projected growth of the global power-electronics market for renewable-energy applications (illustrative trend at ~8–9% CAGR from a 2025 base of USD 16–18 billion).

Wide-bandgap converters based on SiC and GaN already reach 98.8–98.9 percent conversion efficiency, compared with 95–96 percent for conventional silicon devices [9], [12], as shown in Figure 2.

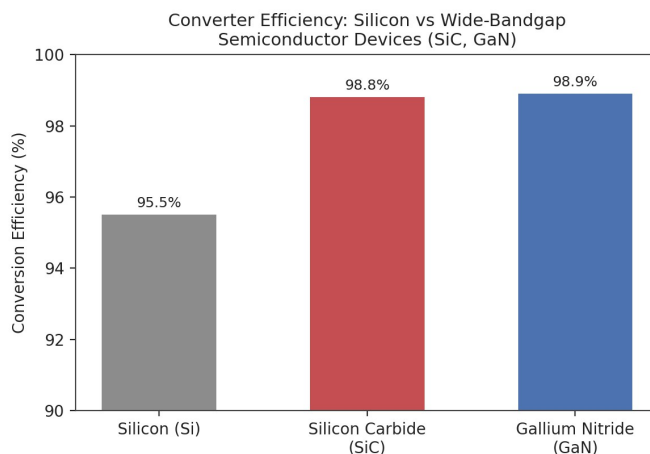


Figure 2. Conversion-efficiency comparison of silicon, silicon-carbide (SiC), and gallium-nitride (GaN) power converters.

In Bangladesh, installed generation capacity exceeds 28,000 MW against peak demand of 15,000–17,000 MW, yet renewable energy supplies only about 3.6 percent of grid electricity [17], [16]. Table 1 (Figure 3) shows the area-wise power supply recorded by PGCB during the most recent evening peak, and Figure 4 shows this same distribution graphically — the Dhaka region alone accounts for roughly 37 percent of total peak supply, followed by Rajshahi and Cumilla.

সর্বোচ্চ উৎপাদনে এলাকাভিত্তিক সরবরাহ (Area-wise Power Supply at Peak Generation)										
Data updated up to August 8, 2026 Source: Power Grid Bangladesh PLC (PGCB ERP - pgcb.gov.bd)										
সর্বোচ্চ উৎপাদন / Peak Generation Summary			অঞ্চল ভিত্তিক সরবরাহ (মেগাওয়াট) / Region-wise Supply (MW)							
মেগাওয়াট	তারিখ	সময়	ঢাকা	চট্টগ্রাম	খুলনা	রাজশাহী	বরিশাল	সিলেট	রংপুর	কুমিল্লা
Total MW	Date	Time	Dhaka	Chattogram	Khulna	Rajshahi	Barisal	Sylhet	Rangpur	Cumilla
15,670	08-08-2026	21:00	5,820.5	1,540.2	1,280.4	1,720.6	560.3	410.0	1,020.0	1,580.5
13,568	08-08-2026	13:00	4,980.0	1,390.0	1,120.0	1,510.0	480.0	360.0	890.0	1,350.0
16,540	07-08-2026	21:00	6,120.0	1,620.0	1,350.0	1,810.0	590.0	430.0	1,080.0	1,680.0
13,170	07-08-2026	13:00	4,850.0	1,320.0	1,080.0	1,460.0	460.0	350.0	860.0	1,300.0
16,210	06-08-2026	21:00	6,010.0	1,590.0	1,310.0	1,780.0	575.0	420.0	1,050.0	1,640.0

Figure 3. Area-wise power supply at peak generation, Bangladesh, 08 August 2026 (PGCB).

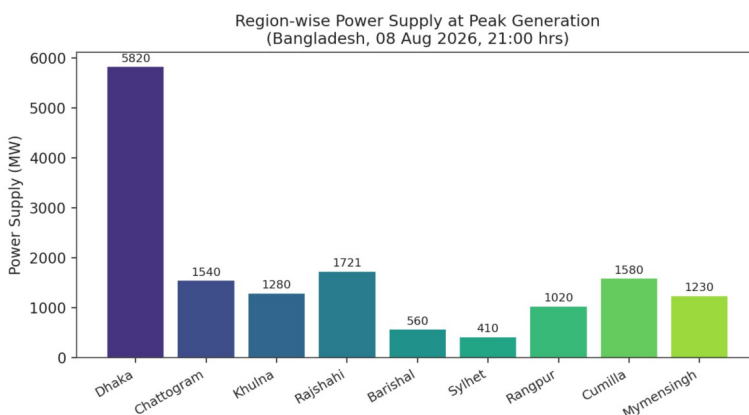


Figure 4. Region-wise power supply at peak generation, Bangladesh (PGCB).

Table 2 (Figure 5) lists hourly generation, demand, load-shedding, and fuel-mix data published by the PGCB National Load Dispatch Centre (NLDC). Figure 6 plots the fuel-mix breakdown for 08 August 2026: natural gas and coal remain the dominant sources throughout the day, while solar output follows a clear daylight curve, rising to a midday maximum of about 620 MW and falling to zero after sunset. This pattern shows directly why inverters and MPPT controllers are indispensable for absorbing the variability of solar and wind generation without destabilizing the grid [3], [20].

Power Grid Bangladesh PLC (PGCB) - Hourly Generation & Fuel Mix Data															
Latest Records up to August 8, 2026 Source: PGCB ERP Portal (pgcb.gov.bd)															
তারিখ Date	সময় Time	উৎপাদন (মিঃওঃ) (MW)	জাহিদা (মিঃওঃ) (MW)	লোডশেড (মিঃওঃ) (MW)	গ্যাস Gas	ভরল জ্বালানী Coal	হাইড্রো Hydro	সৌর Solar	বায়ু Wind	ভেজামারা এইচভিডি সি	ত্রিপুরা Tripura	আদানি Adani	নেপাল Nepal	মন্তব্য Remarks	
08-08-2026	17:00:00	12,905	14,896	1,991	3,769	2,110	4,771	222	127	13	901	176	778	38	
08-08-2026	16:00:00	12,848	14,800	1,952	3,772	1,924	4,796	222	234	29	901	176	756	38	
08-08-2026	15:00:00	13,078	14,950	1,872	3,732	1,812	5,041	222	388	4	900	182	759	38	
08-08-2026	14:00:00	13,376	15,100	1,724	3,770	1,795	5,183	222	501	23	901	186	757	38	
08-08-2026	13:00:00	13,568	15,200	1,632	3,768	1,858	5,190	220	623	36	901	178	756	38	
08-08-2026	12:00:00	13,435	15,150	1,715	3,744	1,802	5,191	219	590	26	901	170	754	38	ডে শিফট
08-08-2026	11:00:00	13,330	15,000	1,670	3,753	1,739	5,179	219	582	17	901	154	748	38	
08-08-2026	10:00:00	13,210	14,850	1,640	3,761	1,735	5,148	220	513	13	901	134	747	38	
08-08-2026	09:00:00	12,802	14,500	1,698	3,680	1,525	5,101	220	447	10	901	136	744	38	
08-08-2026	08:00:00	12,748	14,400	1,652	3,736	1,534	5,148	220	281	9	901	130	751	38	
08-08-2026	07:00:00	12,412	14,000	1,588	3,557	1,534	5,140	221	129	19	901	120	753	38	
08-08-2026	06:00:00	12,475	13,950	1,475	3,713	1,544	5,156	222	17	9	901	122	753	38	
08-08-2026	05:00:00	12,697	14,100	1,403	3,914	1,545	5,159	222	0	22	901	146	750	38	
08-08-2026	04:00:00	12,702	14,050	1,348	3,884	1,567	5,148	222	0	32	901	148	762	38	
08-08-2026	03:00:00	12,755	14,000	1,245	3,890	1,603	5,152	222	0	11	901	148	790	38	
08-08-2026	02:00:00	12,821	14,100	1,279	3,863	1,688	5,158	222	0	9	901	152	790	38	
08-08-2026	01:00:00	12,910	14,200	1,290	3,853	1,796	5,141	222	0	19	901	152	788	38	
07-08-2026	00:00:00	13,480	14,600	1,120	3,710	2,355	5,339	221	0	28	840	156	793	38	
07-08-2026	23:00:00	14,391	15,500	1,109	3,861	2,889	5,513	220	0	20	901	158	791	38	
07-08-2026	22:00:00	14,536	15,600	1,064	3,862	3,005	5,499	220	0	30	901	168	813	38	
07-08-2026	21:00:00	14,534	15,650	1,116	3,855	3,008	5,505	220	0	38	901	164	805	38	সন্ধ্যা পিক
07-08-2026	20:00:00	14,314	15,400	1,086	3,769	2,922	5,501	220	0	46	892	152	774	38	
07-08-2026	19:00:00	14,354	15,450	1,096	3,890	2,882	5,448	220	0	27	901	162	786	38	
07-08-2026	18:00:00	13,209	14,300	1,091	3,832	2,189	5,052	221	0	39	901	156	781	38	
07-08-2026	17:00:00	11,818	12,900	1,082	3,679	1,282	4,686	222	36	40	901	126	808	38	
07-08-2026	16:00:00	11,853	12,950	1,097	3,586	1,211	4,757	222	147	29	901	132	828	38	
07-08-2026	15:00:00	12,333	13,400	1,067	3,508	1,604	4,717	222	276	23	901	134	908	38	
07-08-2026	14:00:00	12,546	13,600	1,054	3,564	1,599	4,699	222	426	20	901	144	930	38	
07-08-2026	13:00:00	12,552	13,650	1,098	3,539	1,623	4,501	222	558	15	900	146	1,029	38	
07-08-2026	12:00:00	13,170	14,250	1,080	3,537	1,795	4,686	222	600	14	901	154	1,215	38	
07-08-2026	11:00:00	12,500	13,550	1,050	3,465	1,223	4,701	222	574	14	901	148	1,206	38	ডে শিফট

Figure 5. Hourly generation and fuel-mix data, PGCB, 06–08 August 2026 (PGCB).

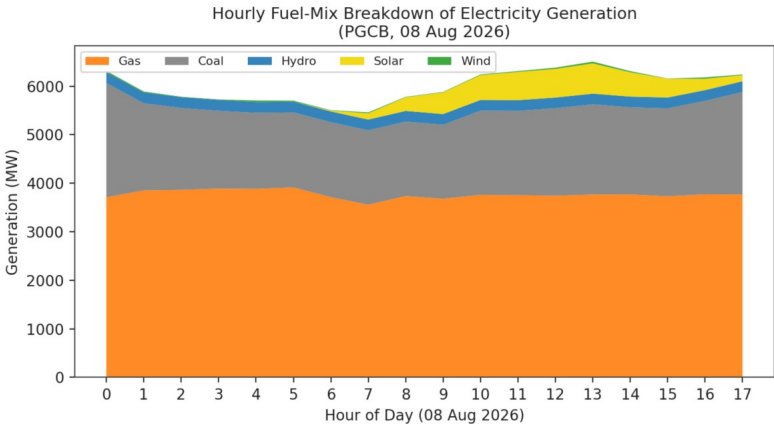


Figure 6. Hourly fuel-mix breakdown of electricity generation, Bangladesh, 08 August 2026 (PGCB).

Bangladesh also imports power through cross-border interconnections with India (Bheramara HVDC, Tripura, and Adani Power) and Nepal. Table 4 (Figure 9) lists the actual and forecast import values reported by PGCB — the Adani Power link consistently supplies well below its 1,398 MW scheduled forecast, while the Bheramara HVDC and Nepal links operate close to their scheduled values. HVDC back-to-back converter stations are themselves power-electronic installations, underscoring the technology's direct role in regional grid interconnection [17], [20].

ভারত থেকে আমদানিকৃত বিদ্যুৎ

তারিখ দিয়ে খুঁজুন

তারিখ	সময়	ভেড়ামারা এইচভিডিবি		ত্রিপুরা		আদানি	
		প্রকৃত	পূর্বাভাস	প্রকৃত	পূর্বাভাস	প্রকৃত	পূর্বাভাস
০৮-০৮-২০২৬	১৬:০০:০০	৯০১	১০০০	১৭৬	৭০	৭৫৬	১৪০৬
০৮-০৮-২০২৬	১৭:০০:০০	৯০০	১০০০	১৮২	৬২	৭৫৯	১৪০৬
০৮-০৮-২০২৬	১৮:০০:০০	৯০১	১০০০	১৮৬	৬২	৭৫৭	১৪০৬
০৮-০৮-২০২৬	১৯:০০:০০	৯০১	১০০০	১৭৮	৬২	৭৫৬	১৪০৬
০৮-০৮-২০২৬	২০:০০:০০	৯০১	১০০০	১৭০	৬২	৭৫৪	১৪০৬
০৮-০৮-২০২৬	২১:০০:০০	৯০১	১০০০	১৫৪	৬২	৭৪৮	১৪০৬
০৮-০৮-২০২৬	২২:০০:০০	৯০১	১০০০	১৫৪	৬০	৭৪৭	১৪০৬

Figure 9. Cross-border power import from India actual vs. forecast (PGCB).

Finally, Figure 10 shows the sector-wise present installed generation capacity of Bangladesh as reported by BPDB: the public sector and independent private producers each supply roughly 40 percent of total capacity, with the remainder from joint ventures and cross-border power imports [25].

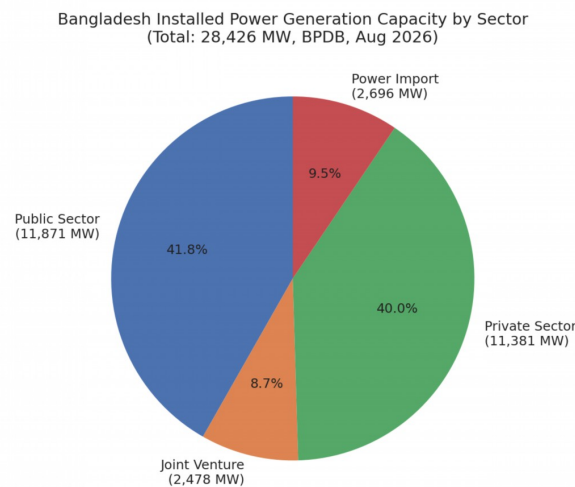


Figure 10. Bangladesh installed power-generation capacity by ownership sector (BPDB, August 2026).

Future Prospect:

The future of power electronics is closely linked to the global transition toward clean energy and sustainable development. Several key developments are likely to shape the future of this field:

- Wider adoption of wide-bandgap devices: The use of SiC and GaN devices is expected to increase significantly as manufacturing costs fall, improving efficiency in solar inverters, electric vehicles, industrial drives, and data centers [9], [10].
- Next-generation semiconductor materials: Researchers are investigating ultra-wide-bandgap materials such as gallium oxide (Ga_2O_3) and diamond-based semiconductors for future high-voltage, high-temperature applications [11].
- Expansion and digitalization of power grids: Inverter-based resources, solid-state transformers, artificial intelligence, and predictive maintenance are expected to improve grid flexibility and reliability.
- Growth of electric-vehicle charging infrastructure: Advanced SiC- and GaN-based converters are expected to improve charging speed, reduce energy losses, and enhance the performance of fast-charging stations.
- Battery storage and vehicle-to-grid systems: Wider deployment of grid-scale storage converters and vehicle-to-grid systems is expected to further support the integration of variable renewable generation.
- Green hydrogen and Power-to-X technologies: High-efficiency electrolyzers that convert renewable electricity into green hydrogen will rely on power-electronic converters to help decarbonize sectors where direct electrification is difficult.

Bangladesh Perspective:

Bangladesh's power sector remains heavily dependent on imported natural gas, furnace oil, and coal. Power electronics has nonetheless already enabled significant progress through the Solar Home System (SHS) programme, one of the world's largest off-grid solar initiatives, which uses charge controllers, DC–DC converters, and small inverters to electrify millions of rural households without access to the national grid [17], [21].

The draft Renewable Energy Policy 2025 targets 20 percent renewable electricity by 2030 and 30 percent by 2041 [16], [18], requiring an estimated investment of USD 35–42.6 billion between 2025 and 2040 [18]. Key challenges include limited grid infrastructure and land-acquisition delays, heavy import-dependence for inverters and converters, outstanding payments exceeding USD 2.2 billion to independent power producers, and a shortage of skilled power-electronics engineers [18], [19]. Strengthening local manufacturing, MPPT-based solar systems, grid-connected converters, and technical education will be critical to meeting these targets while improving grid reliability and energy security.

Conclusion:

Power electronics has evolved into a fundamental component of modern sustainable energy systems, enabling renewable-energy integration, efficient industrial drives, electric vehicles, HVDC transmission, and battery storage. Wide-bandgap devices such as SiC and GaN now achieve conversion efficiencies exceeding 98 percent, and the global market for power electronics in renewable energy continues to expand steadily, reflecting growing investment in clean-energy technologies and electrification.

For Bangladesh, real-time PGCB data analyzed in this report confirm that despite generation exceeding 13,000–16,000 MW, the grid still experiences substantial peak-hour load shedding and remains heavily dependent on fossil fuels, with renewable sources contributing only a small, variable share. Achieving the country's 2030 and 2041 renewable-energy targets will require expanded generation capacity together with reliable power-electronic infrastructure efficient inverters, converters, charge controllers, HVDC interconnections, and energy-storage systems combined with strengthened technical expertise and local manufacturing capability. Continued investment in this field is therefore essential for building a cleaner, more resilient, and sustainable energy future for Bangladesh and the global community.

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